

Interactive Control of Curvature Extrema and Inflections on B-Spline Curves

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Outline

1. Sliding mechanism

Interactive editing under a curvature-extrema bound

2. Closed curves

Ellipse, oval, ovoid, and the general 4-extrema shape space

3. Complex rational B-splines

Möbius transformations

4. PH curves

Lie sphere transformations

Numerators of κ and κ'

For a planar B-spline curve $\mathbf{c}(t) = \sum_i \mathbf{P}_i N_i^d(t)$:

- **Inflections** — zeros of κ , equivalently of $f(t) = \mathbf{c}'(t) \times \mathbf{c}''(t)$ (degree $2d - 3$).
- **Curvature extrema** — zeros of κ' , equivalently of $g(t)$ (degree $4d - 6$):

$$g(t) = \|\mathbf{c}'\|^2 (\mathbf{c}' \times \mathbf{c}''') - 3 (\mathbf{c}' \cdot \mathbf{c}'') (\mathbf{c}' \times \mathbf{c}'')$$

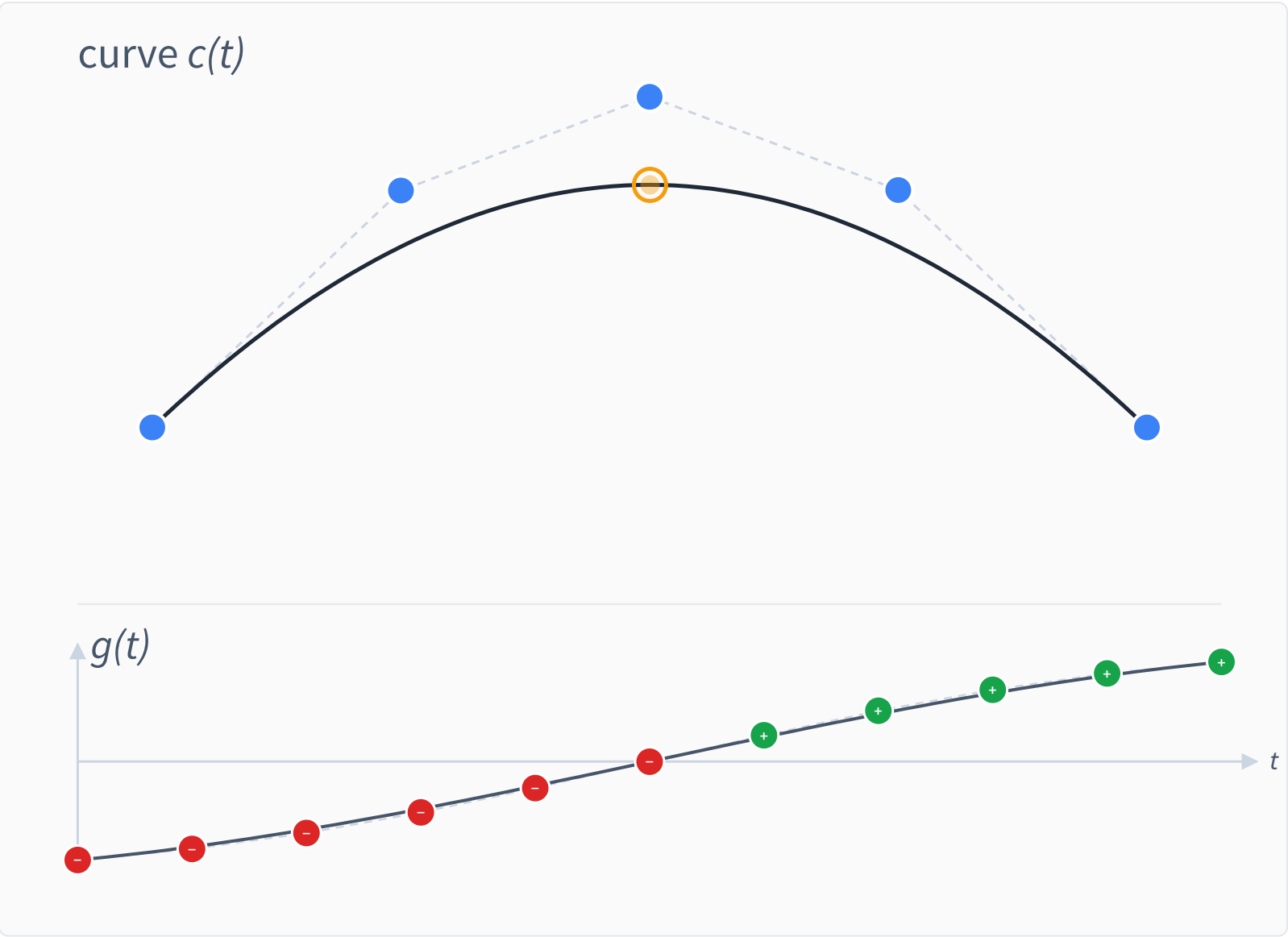
Both f and g are piecewise polynomial — computable exactly in Bernstein form via B-spline algebra.

Without Sliding

Drag any control point. The bound on the number of curvature extrema is preserved — but the sign-change boundary in $g(t)$ is held rigid, so the extremum is **locked in t** .

This is how it worked before the sliding mechanism.

Bound extrema ($S = 1$)

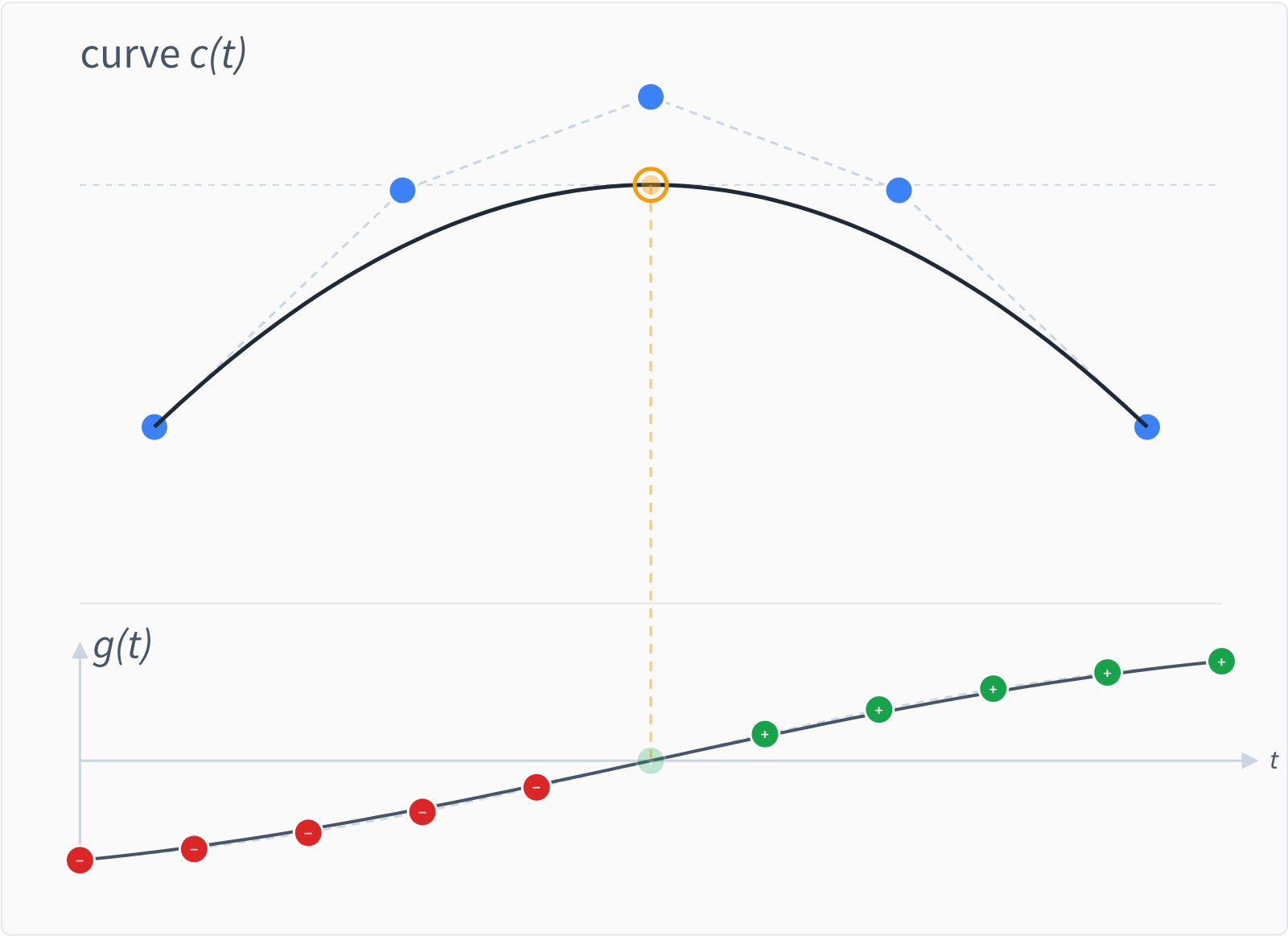


With Sliding

Drag the orange marker — the maximum of curvature.

Below: $g(t)$, the numerator of $\kappa'(t)$ — computable analytically. Its 11 Bernstein coefficients are shown (+ / -); faded dots are inactive constraints, free to slide.

One sign change, moving with the extremum.



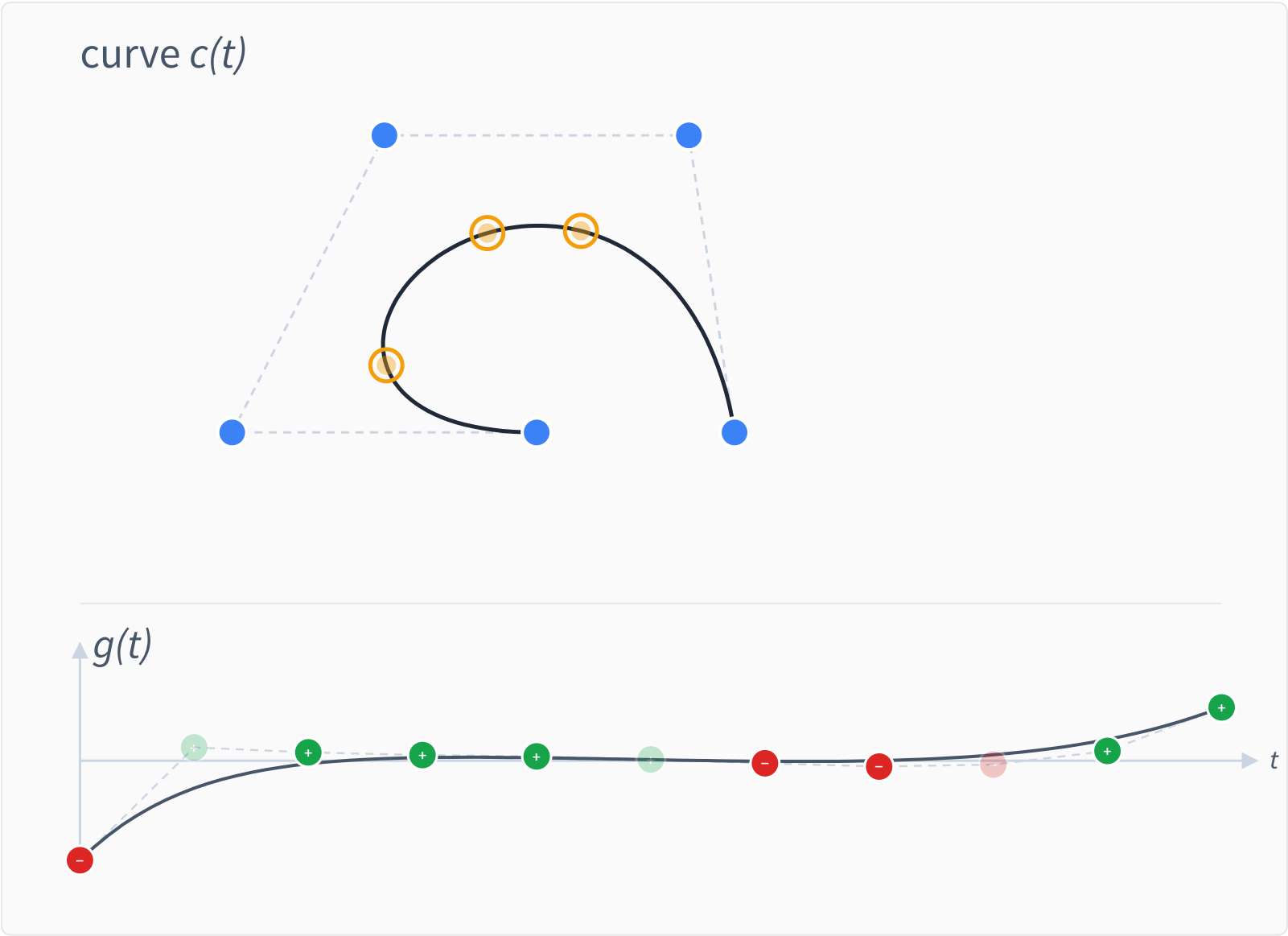
Annihilation

Three curvature extrema at the start. Drag the rightmost control point outward — two extrema slide toward each other and **annihilate**. The bound $S(\mathbf{b})$ drops.

The sliding mechanism makes $S(\mathbf{b})$ *monotone non-increasing*.

Bound extrema ($S = 3$)

Reset



Theorems

Theorem 1 (Schoenberg, variation diminishing). For a B-spline $g = \sum_i b_i N_i$,

$$S^-(g) \leq S^-(\mathbf{b}).$$

Sign changes in the Bernstein coefficients (b_i) bound the number of curvature extrema.

Theorem 2 (contribution). Under constrained editing with the sliding mechanism, $S^-(\mathbf{b})$ is *monotone non-increasing*.

Corollary. The bound on the number of curvature extrema can only stay the same or decrease — across every single edit.

Algorithm

When the user drags control point $\mathbf{P}_k$ toward target $\mathbf{T}_k$, solve:

$$\min_{\mathbf{P}^*} \sum_i w_i \|\mathbf{P}_i^* - \mathbf{T}_i\|^2 \quad \text{s.t.} \quad s_j \cdot g_j(\mathbf{P}^*) \geq 0, \quad j \in \mathcal{A}.$$

The active set $\mathcal{A}$: positions with same-sign neighbors; one **anchor** per alternating sequence — the position with largest $|g|$. Solved by an interior-point method per drag step.

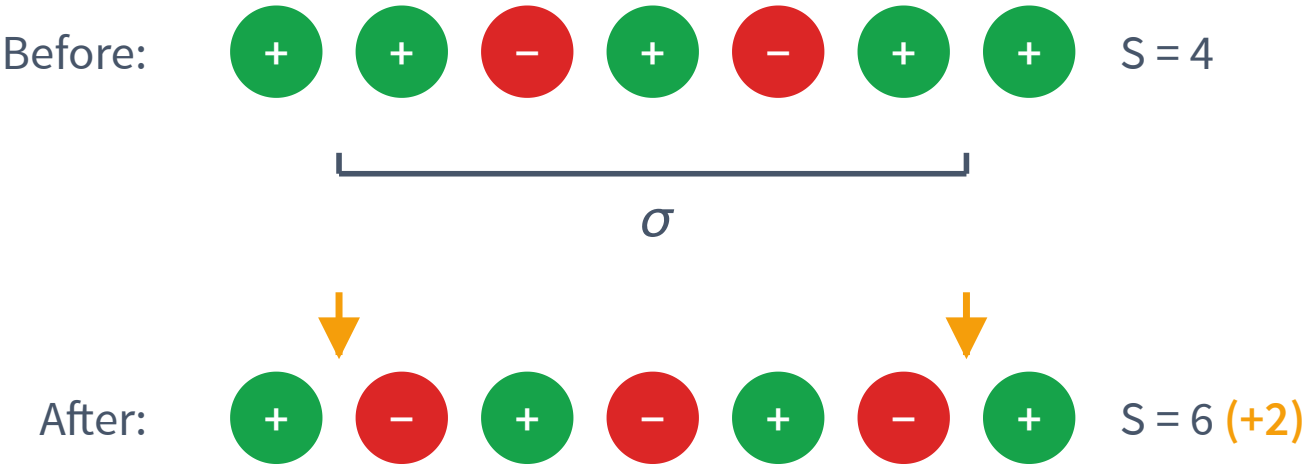
Alternating sequence σ

Inside an alternating sequence σ of $\mathbf{b}$:

- **Interior:** σ already has the *maximum possible* sign-change count (every adjacent pair is a sign change). Any flip can only reduce or preserve interior sign changes.
- **Junctions:** σ 's boundary pairs with the rest of the polygon are *not* sign changes. Each can *become* one if σ 's endpoint flips, at most $+1$ each.

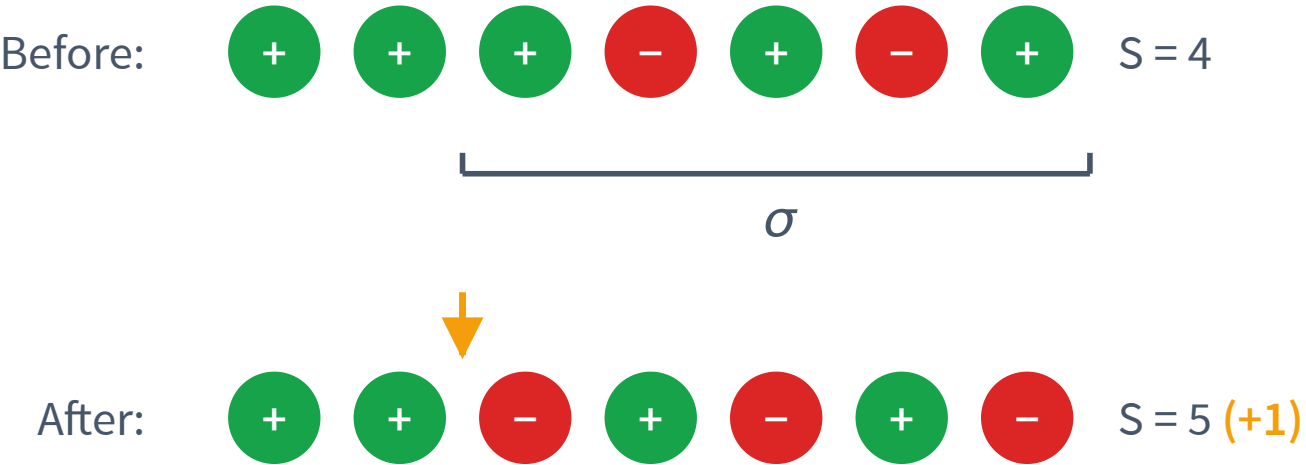
All-flip

Interior σ



Both junctions become new sign changes.

Boundary σ



Only one junction; the other end is the polygon boundary.

*The only flip that grows S^- is all-flip. Excluding any single position of σ from the flip — the **anchor** — blocks the all-flip. S^- does not grow.*

Anchor lemma

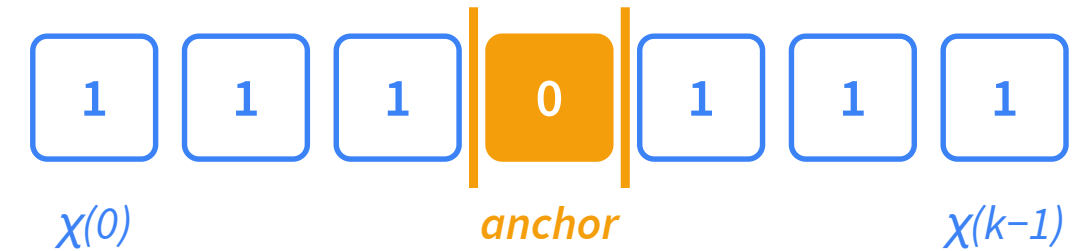
For any flip $G \subseteq \sigma$ that leaves at least one sign fixed (the **anchor**), $S^-(\text{flip}_G \mathbf{b}) \leq S^-(\mathbf{b})$.

Proof. Encode the flip as a boolean pattern χ on σ , 1 if sign is flipped, 0 otherwise.

Each *transition* of χ destroys one interior sign change; $T(\chi)$ counts the transitions.

$$S^-(\text{flip}_G \mathbf{b}) - S^-(\mathbf{b}) \leq \chi(0) + \chi(k-1) - T(\chi).$$

With one anchor the right-hand side is ≤ 0 . ■



Multiple sequences

For a polygon $\mathbf{b}$ with alternating sequences $\sigma_1, \dots, \sigma_n$, **one anchor per sequence** gives safety:
 $S^-(\text{flip } \mathbf{b}) \leq S^-(\mathbf{b})$.

Proof. Apply the anchor lemma to each σ_i independently. A shared junction between adjacent σ_i and σ_{i+1} contributes via XOR:

$$\chi_i(\text{last}) \oplus \chi_{i+1}(0) \leq \chi_i(\text{last}) + \chi_{i+1}(0),$$

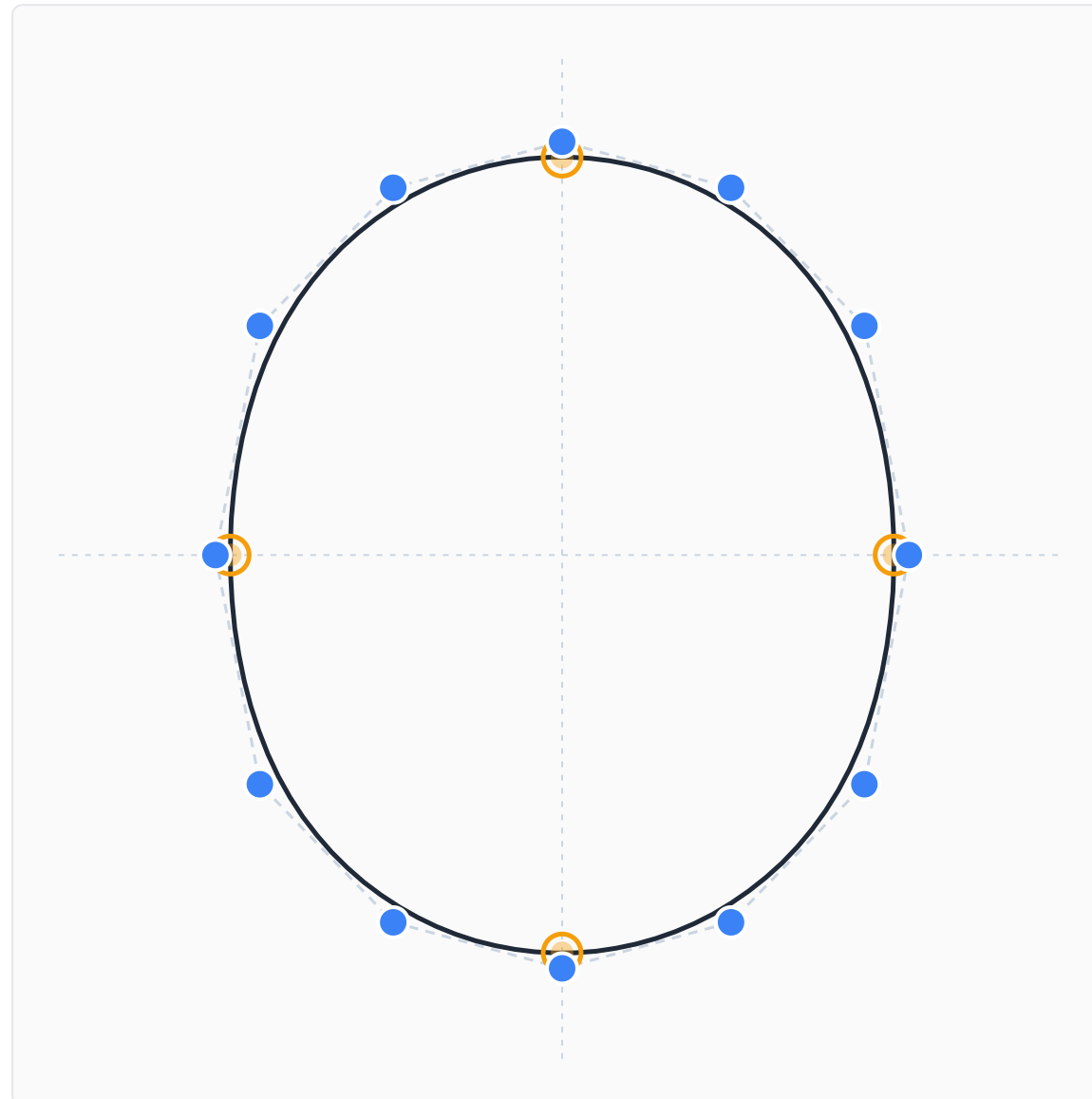
so the per-sequence bounds sum without double-counting. Each is ≤ 0 by the anchor lemma, so the total is ≤ 0 . ■

Oval

A closed curve with

- 2 axes of symmetry
- 4 curvature extrema
- 0 inflections

Drag any control point.

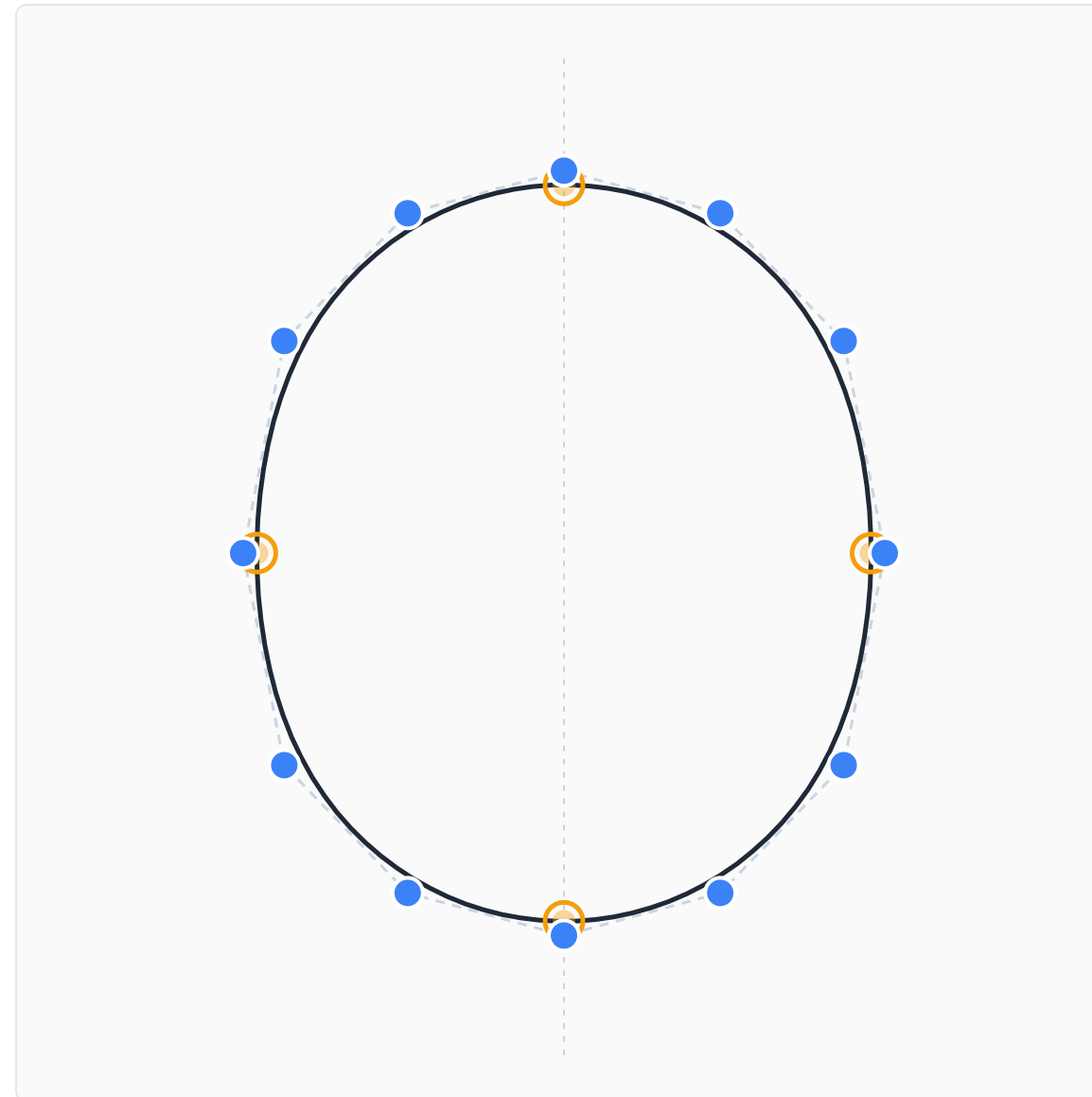


Single axis of symmetry

A closed curve with

- **1** axis of symmetry
- **4** curvature extrema
- inflections *free*

Drag any control point.



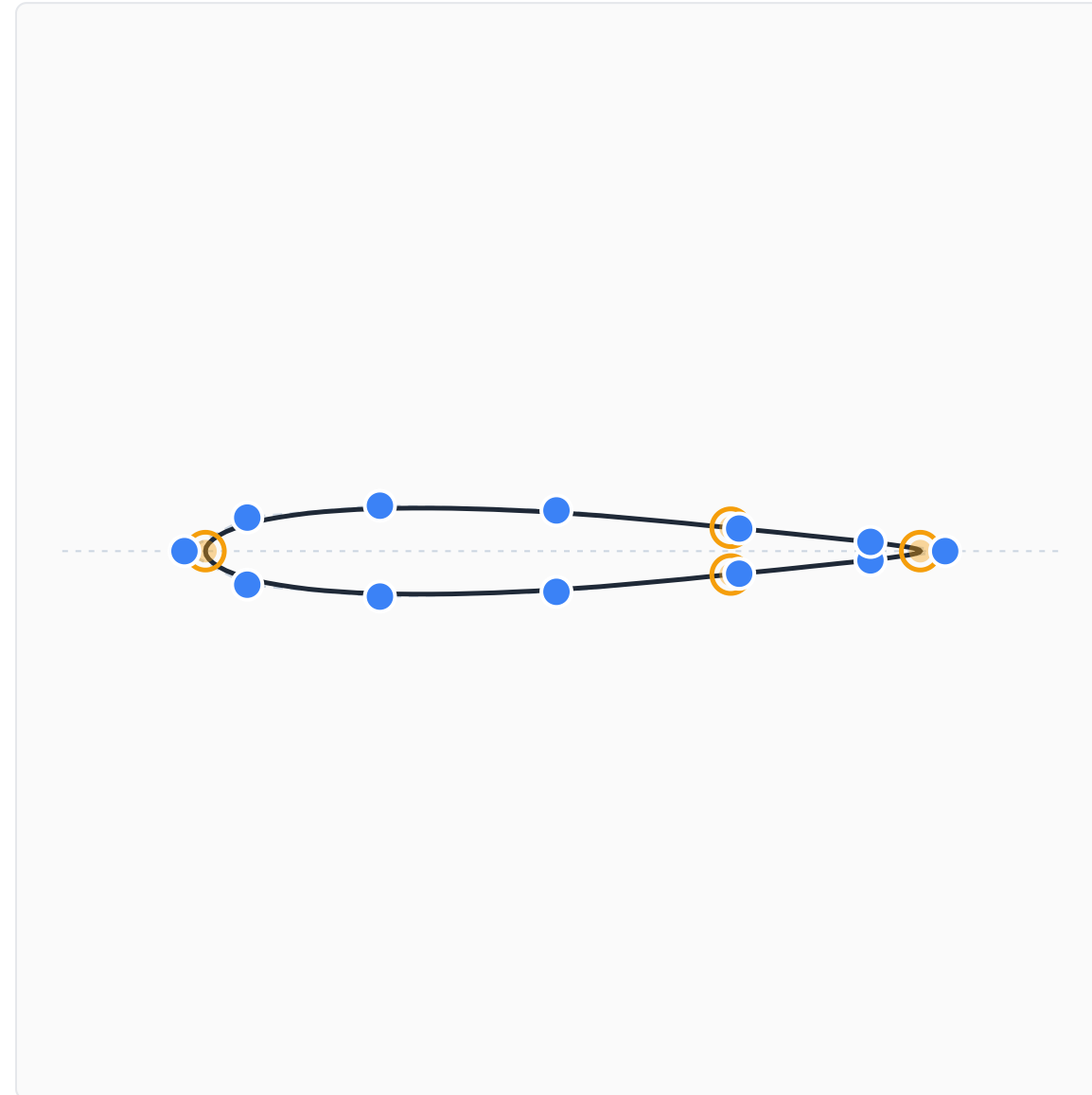
Airfoil

A closed curve with

- 4 curvature extrema
- inflections *free*
- x-axis symmetry —
toggleable

Symmetric

Drag any control point.

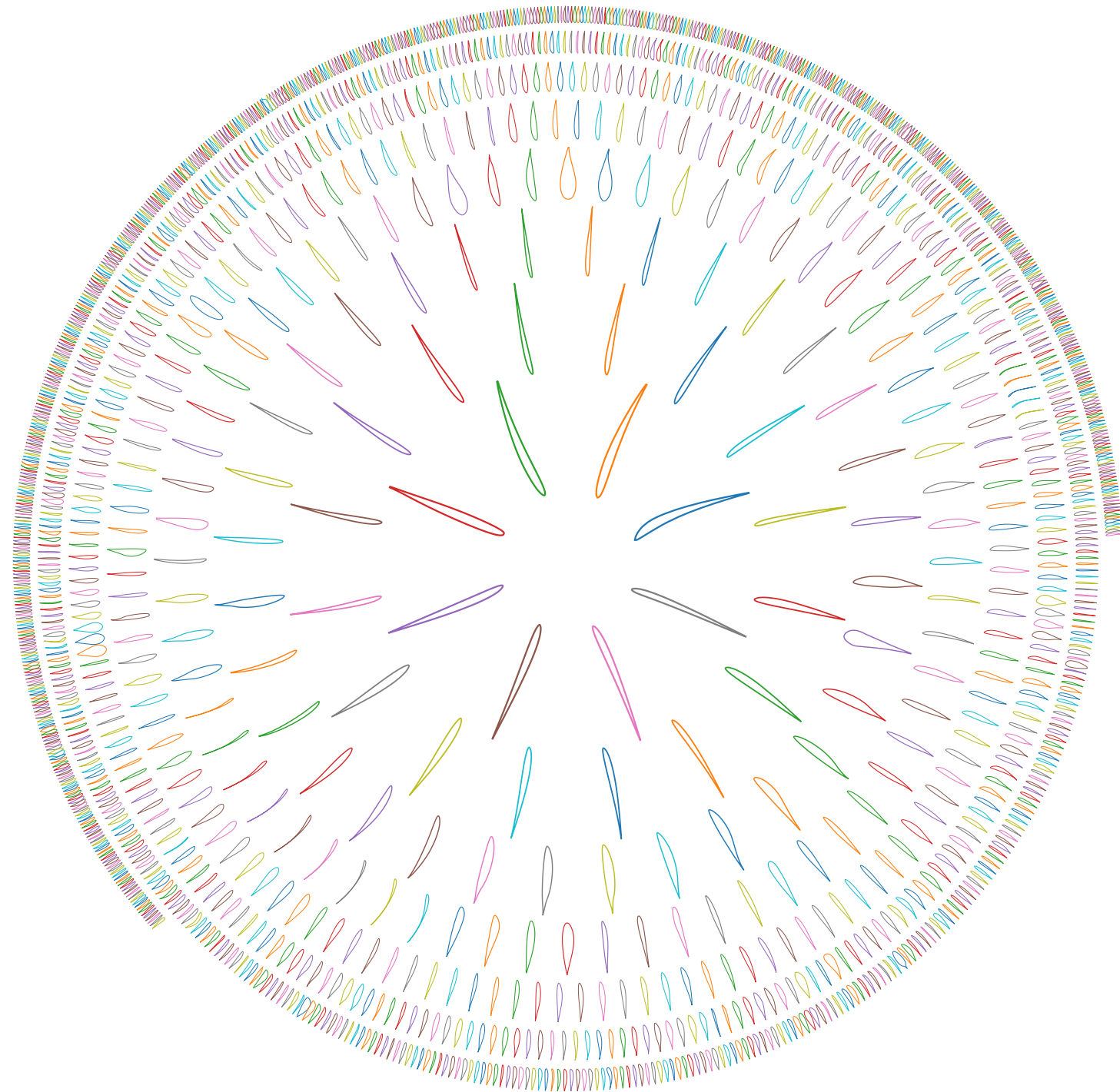


The UIUC airfoil dataset

1644 airfoils, same machinery.

- **98.8%** admit a 4-curvature-extrema fit
- median error **0.13% chord** (p90 = 0.38%)
- **0** errors, **0** fits worsened by the constraint

Workflow: fair $\rightarrow$ fit with an interior-point optimizer $\rightarrow$ a primal-dual solve tightens the fit.

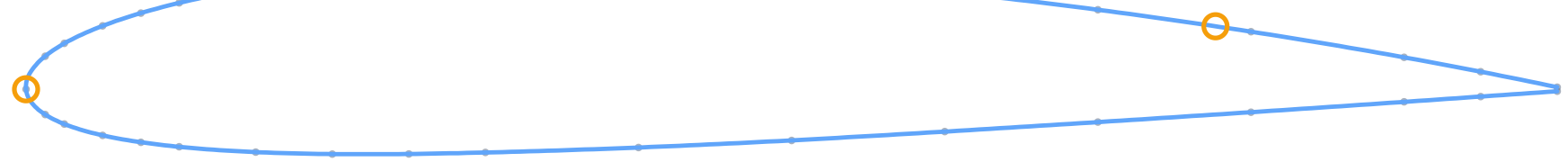


get?

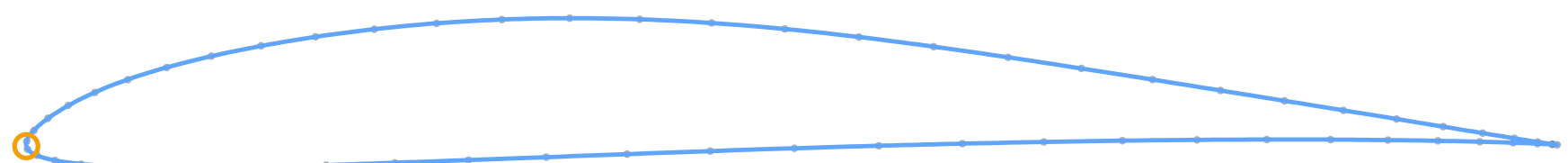
Open curve, trailing edge
pinned — push the
bounded fit to its limit.

- degree-5 B-spline ·
Fair ↔ *Fit* · primal-
dual finish
- 1–2 curvature
extrema, < 0.1%
chord

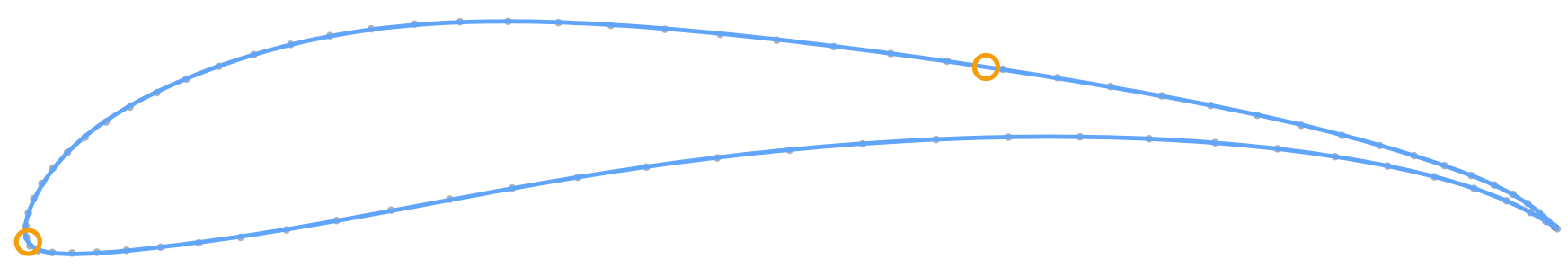
*An AI-assisted algorithm-
design experiment (Claude*



2 curvature
extrema
error
0.008%
chord
Eppler
E387



1 curvature
extremum
error
0.019%
chord
Selig
S1223



(high-lift)
2 curvature
extrema
error
0.071%

Complex Rational B-spline

Move the slider to apply Möbius transformation.

Bound extrema (S = 4)

θ

0°

inv x

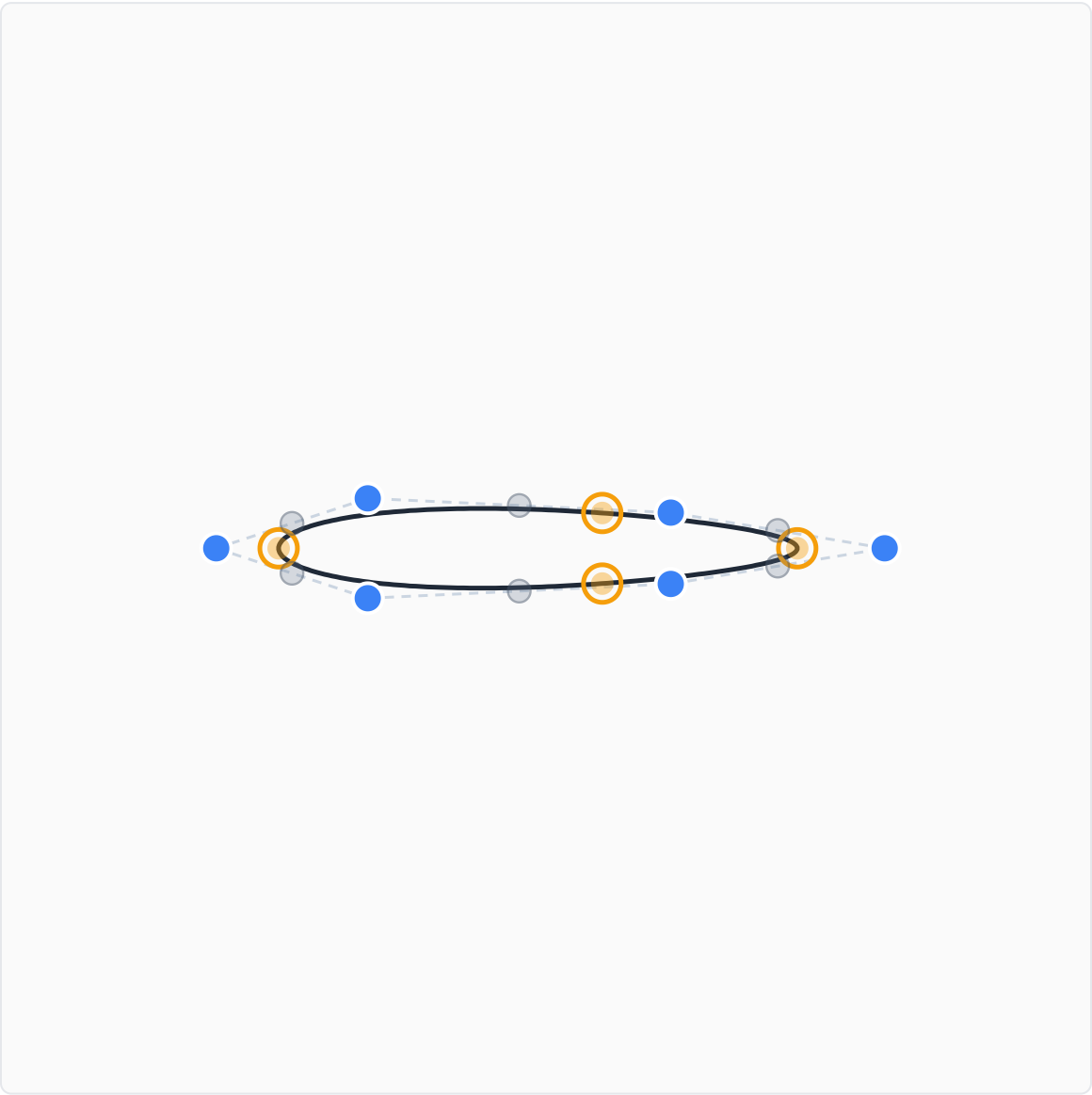
0.00

inv y

0.00

Apply

Reset



PH curves & Lie sphere transformations

A **complex-rational Pythagorean-hodograph** curve $z = A/B$ is built so its hodograph generator is a perfect square — $A'B - AB' = S^2$ — making the speed $|z'| = |S|^2/|B|^2$ rational. So it is PH *by construction*: rational offsets, rational arc length.

- The same κ' sign-change bound applies on the PH curve — edit the generator, the curvature-extrema count is monotone non-increasing.
- Revolve it into a **canal surface**: its **ridges** (curvature extrema of the principal family) are exactly the curve's extrema, swept into rings.
- **Lie sphere transformations** act linearly on the oriented-contact lift and *preserve ridges* — a further bound-preserving editing group, now on the surface.

Conclusion

Sign changes of κ' and κ are bounded — monotone non-increasing under editing.

1. Sliding mechanism + knot insertion
2. Closed curves — the 4-extrema shape space: ellipse $\rightarrow$ oval $\rightarrow$ ovoid $\rightarrow$ general
3. Complex rational B-splines — Möbius transformations
4. PH curves — Lie sphere transformations

References

VARIATION DIMINISHING & SPLINES

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- I.J. Schoenberg, "On spline functions," in O. Shisha (ed.), *Inequalities*, Academic Press, 1967.
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- J.M. Lane, R.F. Riesenfeld, "A geometric proof for the variation diminishing property of B-spline approximation," *J. Approx. Theory* 37 (1983), 1–4.

FAIRNESS & CURVATURE OPTIMIZATION

- N.S. Sapidis (ed.), *Designing Fair Curves and Surfaces: Shape Quality in Geometric Modeling and Computer-Aided Design*. SIAM, 1994. — incl. A.K. Jones, "Curvature integration through constrained optimization" (Ch. 2).
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Thank You

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numericlements.com

Questions?

Backup

The exact sparse Jacobian of $g \cdot$ the PH curves we edit (A, B, S and S, D)

Differentiating with Respect to a Control Point

A B-spline curve is $\mathbf{c}(t) = \sum_j \mathbf{P}_j N_j(t)$.

Differentiate with respect to control point $\mathbf{P}_i$:

$$\frac{\partial \mathbf{c}(t)}{\partial \mathbf{P}_i} = N_i(t)$$

This is itself a B-spline — with the same knot vector and degree, but with control points:

$$0, 0, \dots, 0, \mathbf{1}, 0, \dots, 0$$

A single nonzero control point at position i . Almost entirely zero.

Derivatives Are B-Splines Too

Differentiate $N_i(t)$ with respect to t :

- $N'_i(t)$ is a B-spline of degree $d-1$
- $N''_i(t)$ is a B-spline of degree $d-2$
- $N'''_i(t)$ is a B-spline of degree $d-3$

All have the **same local support** — nonzero on only $d+1$ consecutive spans out of n .

And they depend only on the knot vector and degree, not on the control point positions. Computed **once**, cached for all optimization iterations.

No Need to Multiply Zeros

To compute $\partial g / \partial x_i$, apply the product rule to g . Each term is a product of:

- Intermediate products already computed for g — defined on all n spans, **shared** across all columns
- A cached basis derivative N'_i or N''_i or N'''_i — nonzero on only $d+1$ spans

Multiply only on the $d+1$ nonzero spans. Skip the rest.

For $n = 100, d = 3$: each Jacobian column requires work on **4 spans** instead of 97.

Result: exact Jacobian, **3–5× faster** than automatic differentiation — which does not know what is zero.

The PH curve we edit: A, B, S

Complex-rational Pythagorean-hodograph splines are defined by $z(t) = A(t)/B(t)$, with A, B complex B-splines and a generator S enforcing the *PH condition*:

$$A'B - AB' = S^2$$

The hodograph $z' = S^2/B^2$ is a perfect square, so the square root inside the norm collapses:

$$|z'| = \sqrt{z' \overline{z'}} = \sqrt{\frac{S^2}{B^2} \cdot \frac{\overline{S}^2}{\overline{B}^2}} = \sqrt{\left(\frac{S\overline{S}}{B\overline{B}}\right)^2} = \frac{S\overline{S}}{B\overline{B}}$$

The speed is **rational** — hence rational offsets.

Editing keeps it PH: a constrained solve

Both sides of the PH condition are degree $2p - 2$, so they live in the same **Bernstein basis**.

Equality means matching coefficients — equivalently, the residual must vanish:

$$R = A'B - AB' - S^2 = 0 \quad (\text{every Bernstein coefficient})$$

So editing is a **constrained optimization**, not a formula. The control points A, B, S are *all* free (one weight B_0 pinned as a gauge), and an interior-point solver moves them to:

- **minimize** the distance from the dragged control point to the cursor,
- **subject to** $R = 0$ (stays a PH curve) and the sign constraints $s_j g_j \geq 0$ (keeps the curvature-extrema count).

Letting S float alongside A, B is what lets the curve bend, rotate and scale freely while never leaving the PH family.

Or: PH by construction — the (S, D) form

A second route drops the constraint entirely. Write $z = F/D$ and choose the **generator** S and denominator D as the free variables. The hodograph numerator is fixed to a square by *fiat*:

$$H = S^2 \implies F'D - FD' = S^2 \text{ (Wronskian)}$$

This is **linear in F** , so the numerator is simply *recovered* — by integration when D is constant ($F = \int (S/D)^2$), or a small per-span linear solve otherwise.